Steady-State Model and Power Flow Analysis of Electronically-Coupled Distributed Resource Units

Hassan Nikkhajoei[1], Reza Iravani[2]

[1]University of Wisconsin-Madison, [2]University of Toronto

This paper presents closed-form steady-state, fundamental-frequency models of power electronic converter systems for coupling Distributed Resource (DR) units to the utility power grid. Based on the developed models, the paper introduces a two-step power flow analysis approach that accurately and efficiently represents electronically-coupled DR units. A feature of the proposed approach is that it solves for the internal variables of each DR unit in a non-iterative fashion based on the presented closed-form DR models, and thus, enhances efficiency and accuracy of the solution. This papers provides the details for steady-state closed-form formulation of matrix converter and voltage-sourced AC-DC-AC converter for coupling DR units. However, the presented methodology is conceptually applicable to other types of converter systems. An application of the proposed power flow approach to a micro-grid that includes two electronically-coupled DR units is also presented.